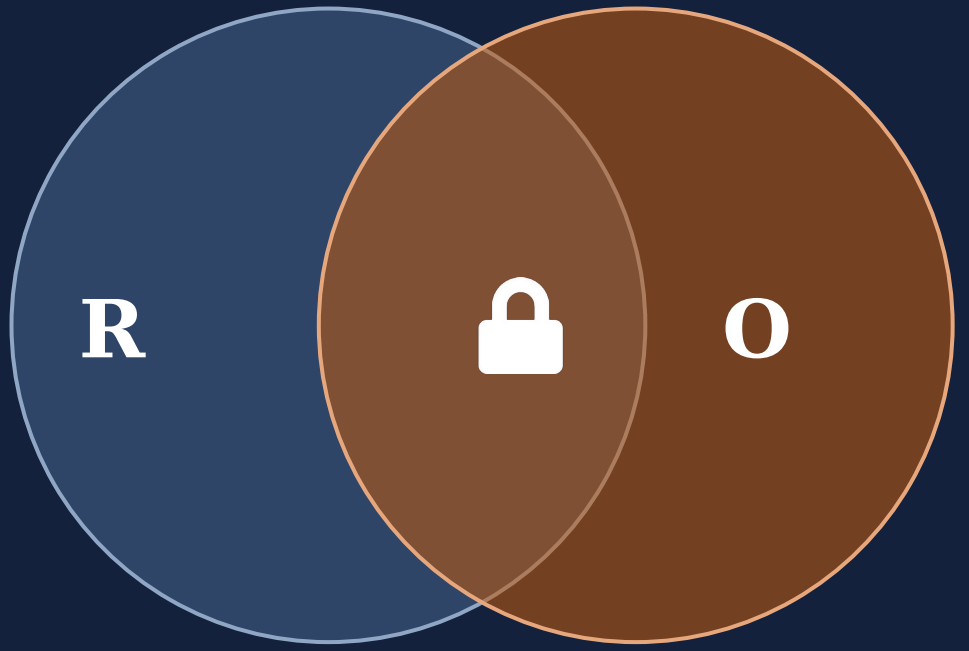
Privacy-Preserving Vertical Federated Linear Regression

When the organization holds the response: protocol, privacy analysis and evaluation



Working draft · September 2026

1

The problem

Joint regression across data that cannot be pooled

Two datasets, one question, no pooling



Researcher R

- Smaller cohort with rich domain variables: health, education, surveys
- Wants to know whether these predict an administrative outcome
- Holds features X_R and identifiers



Organization O (e.g. StatCan)

- Population register: tax filings, income, benefits
- Holds the outcome y and features X_O
- Cannot release records; law and governance bar direct sharing

6–8 weeks of vetting and approvals today before a researcher can link a cohort to agency records in a Research Data Centre. Neither dataset alone answers the question.

Same people, different variables

Vertically partitioned data: overlapping rows, disjoint columns



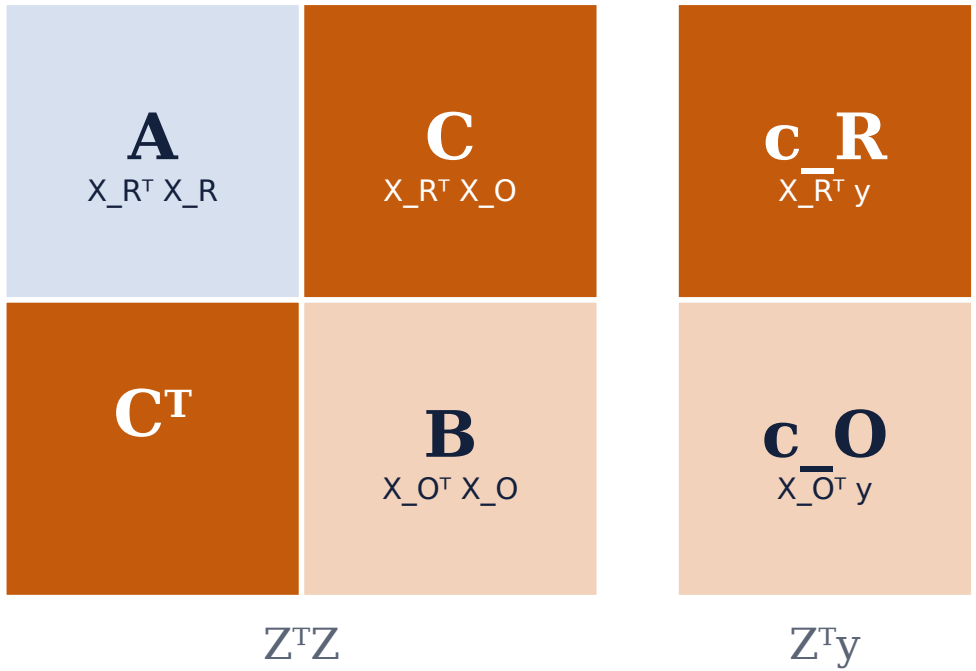
Goal

- Fit y on [X_R | X_O] over the n matched individuals
- Neither party sees the other's raw data
- O must not learn which of its records are in R's cohort
- The released model must not expose individuals

What has to be computed

OLS on the matched join needs a Gram matrix that spans both parties

$$\beta = (Z^T Z)^{-1} Z^T y, \quad Z = [\, X_R^I \mid X_O^I \,]$$



R alone: exact, never leaves R

O's data, but only on matched rows (needs b)

Cross-party: needs both parties' data

Only A is purely R's. Everything that touches O's data is computed under R's encryption key and noised by O before R sees it.

Three guarantees, honest-but-curious parties

Input privacy

Each party's view can be simulated from its own input and the agreed output. Neither sees the other's raw data.

Linkage privacy

O learns nothing about which of its records match R's cohort (only public sizes).

Output privacy

What R receives is differentially private for each matched person's O-side data (x_O, y), with R's features held fixed (replace-one).

Parties follow the protocol but analyse everything they receive. R holds all decryption keys and learns the match set by design; O controls the privacy budget.

Where this work sits

Approach	Where the data sit	Record linkage	Output privacy	Gap for our setting
Central DP regression (AdaSSP, Wang 2018)	One trusted curator holds the joined table	Not needed	DP	Assumes the pooled data exist
VFL with HE (e.g. Yang et al. 2019)	Split across parties	Assumed or separate PSI	Usually none	Model can leak individuals
Private set intersection (Chen–Laine–Rindal)	Split across parties	Yes	Not applicable	Linkage only, no model
This work	Split; never pooled	Labeled PSI, private to O	DP, budget controlled by O	Narrower guarantee: protects O's side; R learns membership

Same statistical engine as AdaSSP (perturb the sufficient statistics, then solve a ridge), delivered without pooling, and with R's own block kept exact.

2

What we did

Reviewed the draft, found what breaks, and rebuilt it



1. Review

Draft paper and code checked line by line

2. Run

End-to-end over sockets: plaintext and real OpenFHE

3. Attack

Three privacy reviews plus an independent critique

4. Redesign

Revised protocol, TikZ figure, rewritten paper

5. Simulate

Accuracy and bias-correction studies

6. Benchmark

AdaSSP on 29 of Wang's UCI datasets

Critical finding: the noise did not protect the ciphertext

100%

of trials in which R identified the true dataset from O's reply: matched and unmatched records alike (real OpenFHE, 20/20).

73%

the most that ε = 1 differential privacy allows.

Why

- The draft added the noise as a plaintext scalar, which changes only c_0
- c_1 is a deterministic function of R's own ciphertexts, keys and O's data
- R, who holds the key, recomputes c_1 for each candidate dataset and compares

Fix

- Add a fresh encryption of the noise, which re-randomizes c_1
- Switch to the last modulus level, then flood the error term
- BFV linkage replies: statistical flooding; CKKS: DP-calibrated flooding

Review scorecard

Finding	Severity	Status in the revised protocol
Plaintext-scalar noise leaks through the ciphertext	**Critical**	Fixed: fresh encryption plus flooding
Calibration step shares raw-data statistics	**Critical**	Fixed: per-party commitments, public constants only
Add/remove sensitivity vs replace-one adjacency (true ε up to ~2× the claim)	**High**	Fixed: replace-one Δ, closed form
No privacy ledger across runs	**High**	Fixed: zCDP ledger at O
PSI reveals O's row order and bin loads	**High**	Fixed: secret permutation, public padding
Adaptive λ cancels the bias correction	**Validity**	Fixed: λ set from public inputs
CKKS slot leak; HE parameter security	**Checked**	Not an issue with OpenFHE
R learns which of its cohort are in the register	**By design**	Stated in Definition 4

Contributions

1 Corrected data flow

R's Gram block A is computed locally, is never sent and never noised. The sensitivity loses its B_R^4 term.

2 Revised protocol

Sanitized replies, replace-one DP, zCDP ledger, per-party calibration, hardened PSI, ridge fixed from public inputs.

3 Theory

Finite-noise bias as a conditional expectation, a zero-budget correction, and a closed-form replace-one sensitivity.

4 Evaluation

End-to-end runs, simulation studies of accuracy and of the correction, and a benchmark against AdaSSP.

3

Methodology

The revised protocol, phase by phase

The protocol at a glance

0 **Setup and commitments**

Only public constants cross

1 **Private linkage**

BFV labeled PSI; R learns the match set

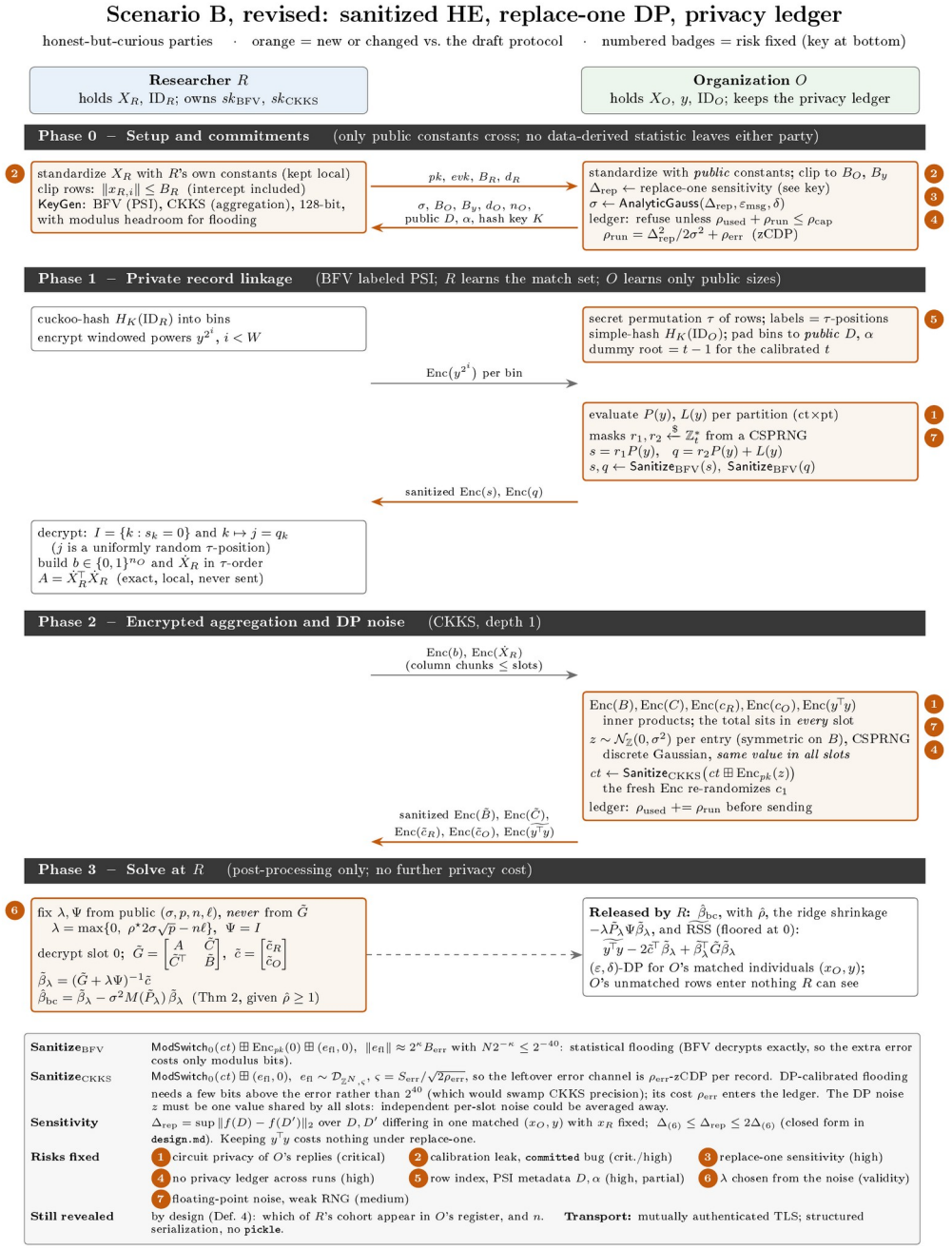
2 **Encrypted aggregation**

CKKS, depth 1; DP noise added by O

3 **Solve at R**

Fixed ridge, bias correction; post-processing

Orange in the figure marks what changed from the draft.



Phases 0–1: commitments and private linkage

0 Setup and commitments

- R: standardizes with its own constants (kept), clips rows to B_R, generates BFV and CKKS keys
- O: standardizes with public constants, clips to B_O and B_y
- O computes Δ_{rep} and σ, and refuses the run if the ledger would exceed its cap
- **Only public constants cross: no quantiles, clip rates or sample moments**

1 Labeled PSI with alignment

- O applies a secret permutation τ, so labels are random positions
- Keyed hashing; bins padded to a public degree D and count α
- R cuckoo-hashes its IDs and sends encrypted windowed powers
- O evaluates membership and label polynomials, masks, and sanitizes
- R learns I and the aligned positions; O learns only public sizes

Phase 2: encrypted aggregation

O multiplies R's encrypted vectors by its own plaintext columns (depth 1)

$$\mathrm{Enc}(B_{jk}) = \Sigma_i \mathrm{Enc}(b_i) \cdot X_{O,ij} \cdot X_{O,ik}$$

$$\mathrm{Enc}(C_{jk}) = \Sigma_i \mathrm{Enc}(\dot{X}_{R,ij}) \cdot X_{O,ik}$$

$$\mathrm{Enc}(c_{O,k}) = \Sigma_i \mathrm{Enc}(b_i) \cdot X_{O,ik} \cdot y_i$$

$$\mathrm{Enc}(c_{R,j}) = \Sigma_i \mathrm{Enc}(\dot{X}_{R,ij}) \cdot y_i$$

b: selection vector; $\dot{X}_R$: R's features in O's order (zero off the match set)

$A = \dot{X}_R^T \dot{X}_R$ stays with R

Exact, never transmitted, never noised.

One noise draw per entry

z is the same in every CKKS slot. Independent noise per slot could be averaged away.

Each released entry: $ct \leftarrow \mathrm{Sanitize_CKKS}(ct \boxplus \mathrm{Enc}_{pk}(z))$, $z \sim$ discrete Gaussian $N_{\mathbb{Z}}(0, \sigma^2)$ from a CSPRNG

Calibrating the noise, and keeping count

Replace-one sensitivity

The adjacency replaces one matched person's (x_O, y). The draft used the add/remove bound, which understates Δ.

(B_R, B_O, B_y)	Δ_add	Δ_rep	Ratio
(1, 1, 1)	2.24	3.16	1.41
(3.16, 3, 3)	20.6	29.4	1.43
(√5, √5, 5)	30.4	31.6	1.04
Simulation population	58.2	86.2	1.48

$\Delta_{add} \le \Delta_{rep} \le 2\Delta_{add}$. Releasing $y^\top y$ costs nothing under replace-one.

σ from the analytic Gaussian

Smallest σ meeting (ε, δ) for the whole stacked release (Balle–Wang). Discrete Gaussian at the CKKS resolution.

zCDP ledger at O

$\rho_{run} = \Delta^2/2\sigma^2 + \rho_{err}$. O refuses a run once the cumulative spend would pass its cap. Re-runs, retraining and inference queries all count.

Sanitizing O's replies (circuit privacy)

R holds the key, so every reply must reveal nothing beyond its decrypted value

1 Fresh encryption

ct ⊞ Enc_pk(z) makes c_1 computationally uniform: it no longer depends on O's data.

2 Modulus switch

Drop to the last level, which scales down the data-dependent error.

3 Flood the error

BFV: statistical flooding with N·2^−κ ≤ 2^−40. CKKS: DP-calibrated flooding with width ς = S_err / √(2ρ_err).

Why DP-calibrated flooding for CKKS

Classical 2^{40} flooding would swamp CKKS precision. A DP-sized flood needs only a few bits above the error; its cost joins the ledger.

Open proof obligation

An explicit bound on S_err, how much one record can change the ciphertext error, for the exact encoding and key switching.

Phase 3: fixed ridge and zero-budget bias correction

Ridge, fixed before decryption

$$\lambda = \max\{\, 0,\, 2\rho^{*}\sigma\sqrt{p} - n\ell \,\}, \quad \Psi = I$$

Release gate

$$\hat{\rho} = \lambda_{\min}(\tilde{G} + \lambda I) \,/\, (2\sigma\sqrt{p}) \geq 1$$

Private ridge estimate

$$\beta_{\lambda} = (\tilde{G} + \lambda I)^{-1}\, \tilde{c}$$

Bias-corrected estimate

$$\beta_{bc} = \beta_{\lambda} - \sigma^{2}\, M(\tilde{P}_{\lambda})\, \beta_{\lambda}$$

Theorem (conditional bias)

$$E[\, \beta \mid \mathscr{E}_{\eta} \,] - \beta = \sigma^{2}\, M(P)\, \beta + O(\sigma^{4}) + O(\exp(-p(\eta\rho - 1)^{2}/2))$$

$$M(P) = (P^{2} + \mathrm{tr}(P)P - P\,\mathrm{diag}(P))\,\Pi_{O} + (P\,\Pi_{O}\,P + \mathrm{tr}(\Pi_{O}\,P)\,P)\,\Pi_{R}$$

The plain mean does not exist, hence the conditioning on $\mathscr{E}_{\eta}$.

Why λ must not depend on the noise

Chosen from $\tilde{G}$, λ moves with the noise. The correction then removed none of the bias (1.0–1.6× left); with fixed λ it removes 87–98%.

How we evaluate

Simulation population

Features on real-world scales (income, age, binary indicators, counts), correlated. Public standardization constants; committed 99.5% bounds; $R^2 \in \{0.3, 0.5, 0.9\}$.

Monte Carlo

n from 100 to 100k matched records; $\varepsilon \in \{0.5, 1, 2, 4, 8\}$, $\delta = 10^{-5}$; 200 replicates, each with new data and a new DP release.

Metrics

RMSE against the true β (standardized); per-coefficient efficiency ratio; interval coverage; MSE ratio corrected/uncorrected.

Benchmark

29 UCI datasets from Wang's published run, with his preprocessing; AdaSSP as published and under our guarantee; 10-fold test MSE.

4

Results

What we have so far, and what is still to run

The cryptographic layer is exact

Draft protocol, run end-to-end over sockets with plaintext and real OpenFHE backends

0

misaligned rows

PSI recovered exactly the true match set in all 6 cohorts, plaintext and OpenFHE

$\approx 10^{-10}$

max |β − β_OLS|

with the noise switched off: encryption adds no error

≈13 s

end-to-end on OpenFHE

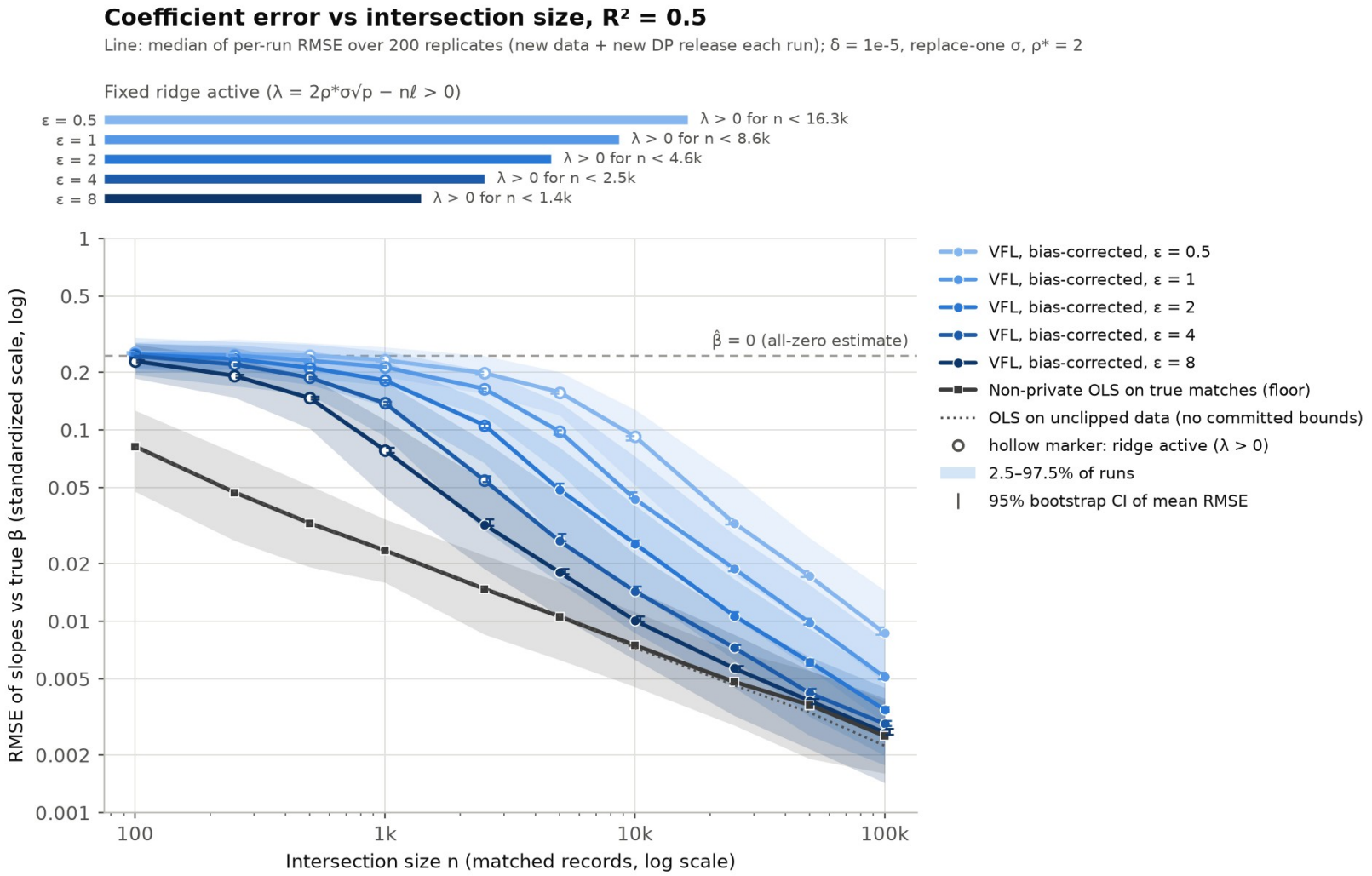
n = 400 matched out of n_O = 4,000 records (OpenFHE 1.5.1)

Cost of the revised protocol in OpenFHE

To be run

Register size n_O	d	PSI time (s)	Aggregation time (s)	Traffic (MB)	Sanitization overhead
4,000	10	*tbd*	*tbd*	*tbd*	*tbd*
32,000	10	*tbd*	*tbd*	*tbd*	*tbd*
65,536	20	*tbd*	*tbd*	*tbd*	*tbd*
10^6 (chunked)	20	*tbd*	*tbd*	*tbd*	*tbd*

Accuracy against intersection size



Within 2× of non-private OLS at

ε = 8	**n ≈ 3.2k**
ε = 4	**n ≈ 9.4k**
ε = 2	**n ≈ 34k**
ε = 1	**n ≈ 105k**

- Small n: the ridge shrinks $\hat{\beta}$ towards 0
- Then the DP error falls like 1/n and merges into the OLS floor
- Run-to-run spread is about ×2.5–3

Key numbers

Median RMSE of the standardized slopes against the true β (ratio to the OLS floor), $R^2 = 0.5$

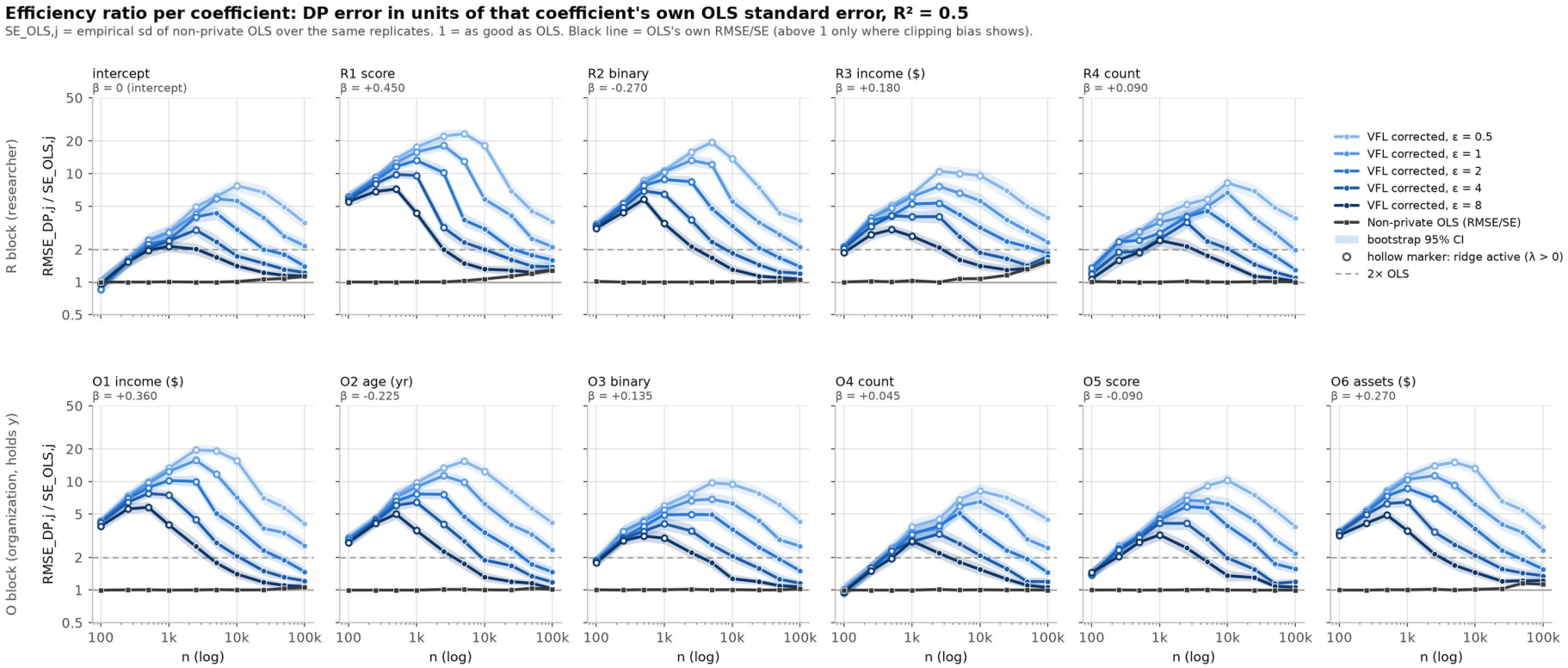
n	ε = 0.5	ε = 1	ε = 2	ε = 4	ε = 8	OLS floor
1,000	0.231 (9.9×)	0.213 (9.1×)	0.181 (7.7×)	0.138 (5.9×)	0.078 (3.3×)	0.023
10,000	0.092 (12×)	0.044 (5.8×)	0.025 (3.4×)	0.014 (1.9×)	0.010 (1.3×)	0.0075
100,000	0.0087 (3.4×)	0.0051 (2.0×)	0.0034 (1.4×)	0.0029 (1.2×)	0.0026 (1.0×)	0.0025

n at which VFL comes within 2× of the OLS floor

R^2	ε = 0.5	ε = 1	ε = 2	ε = 4	ε = 8
0.3	≈240k*	75k	21k	7.1k	1.9k
0.5	≈380k*	≈105k*	34k	9.4k	3.2k
0.9	>100k	>100k	66k	30k	13k

** extrapolated beyond n = 100k*

Comparing coefficients on different scales



Raw scale

Coefficients span 5.6 orders of magnitude (≈$0.2 per $ income vs ≈$10k per binary). A pooled raw RMSE is 87% driven by two binary coefficients.

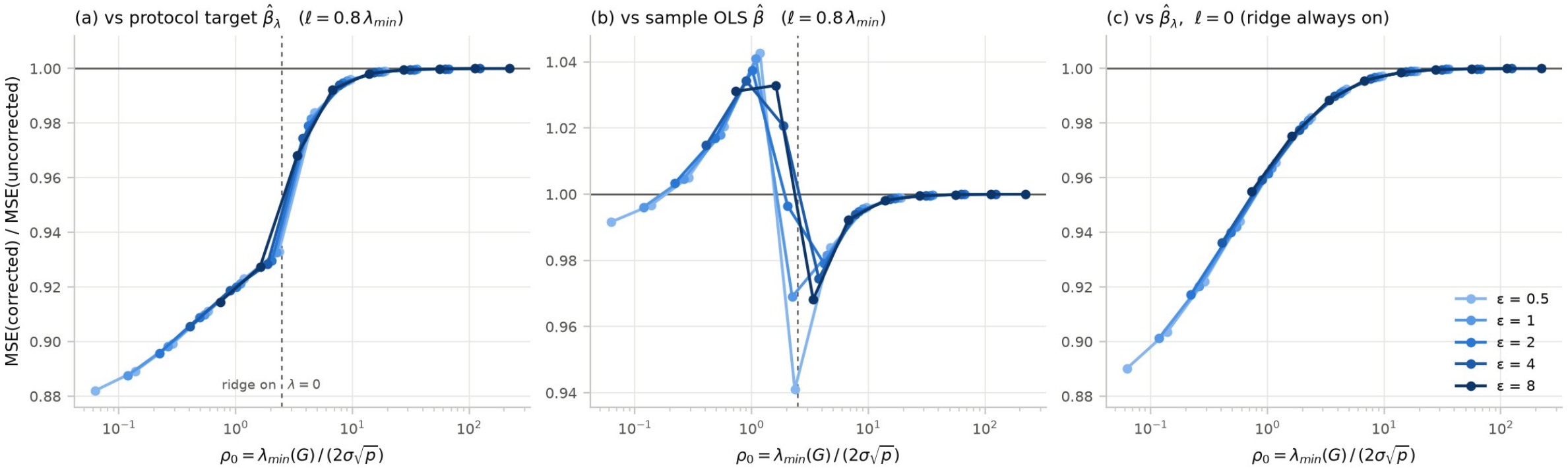
Efficiency ratio

RMSE_DP,j / SE_OLS,j: unit-free. A ratio r means the release is about as accurate as OLS on n/r^2 records.

Result

Once the ridge is off, DP costs about the same for every coefficient (±15%); O-side coefficients run ~10% higher.

Does the bias correction matter?



−7 to −12%

MSE vs the protocol target when the ridge is on. The correction also shrinks noise; it never raised this MSE (90 settings).

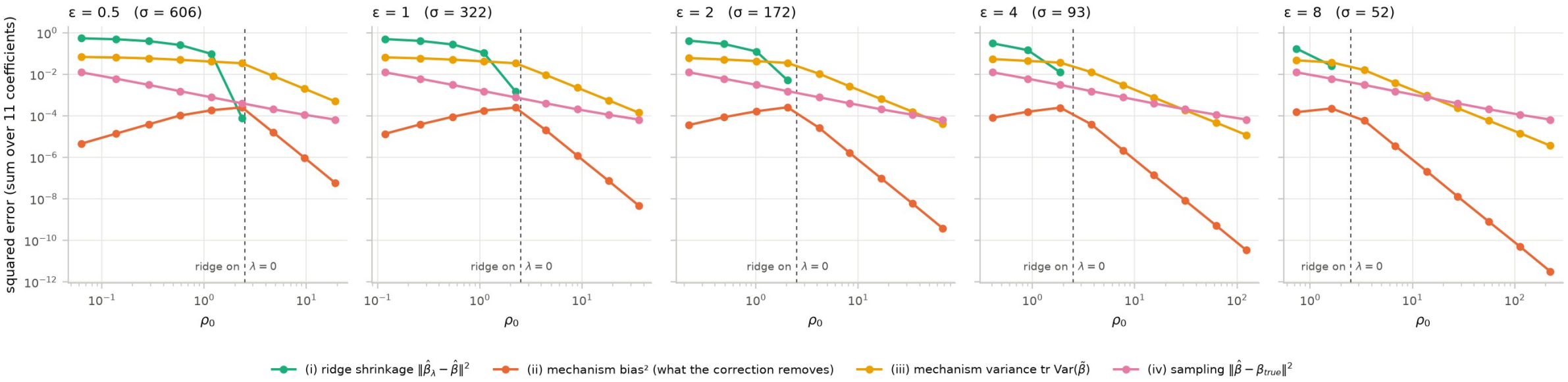
+0.3 to +4.3%

MSE vs OLS under a moderate ridge: the correction restores shrinkage that noise had offset.

≥ 130

averaged releases before the bias alone would matter; one release: at most 0.75% of MSE.

Where the error comes from when σ is large



Error anatomy

- Heavy ridge: 72–90% of the error is shrinkage, 11–31% noise variance
- Mechanism bias²: at most 0.75% anywhere

Inference

- λ = 0: 95% intervals cover 0.94–0.96; SEs within 1–3% of the truth
- Ridge on: coverage collapses. Clipping shifts the target at large n (every DP method pays this)

Benchmark against AdaSSP (Wang 2018)

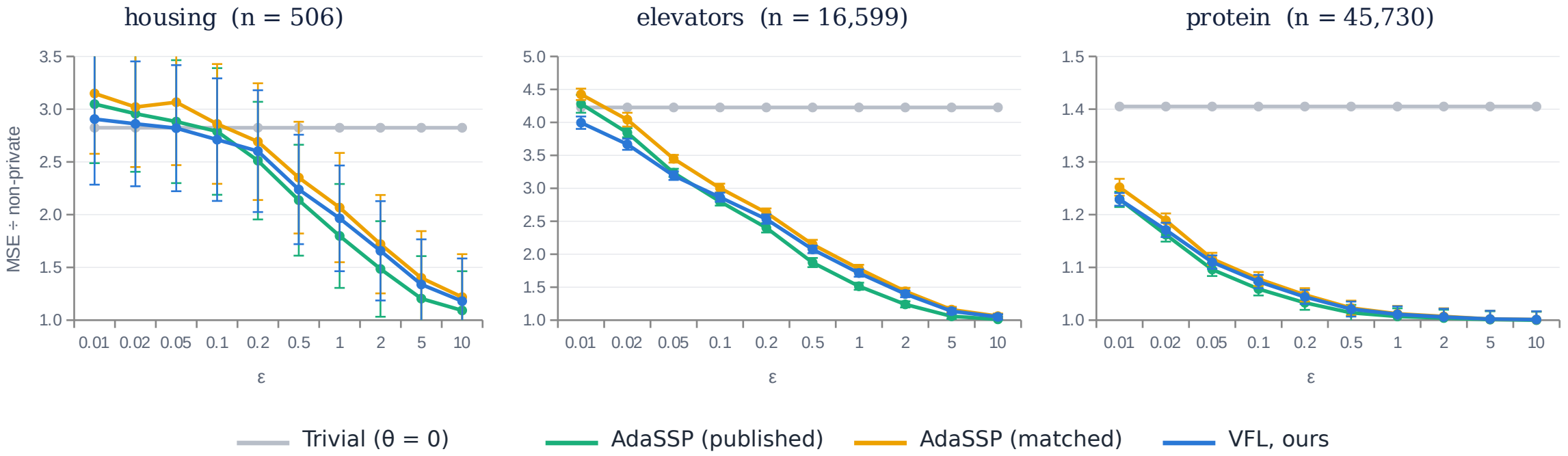
Test MSE at $\varepsilon = 1$, $\delta = 10^{-6}$, 10-fold CV. Ratio to non-private in brackets; bold = best private method

Dataset	n	d	Non-private	AdaSSP (published)	AdaSSP (matched)	VFL, ours	VFL, uncorrected
housing	506	13	0.0396	**0.0713 (1.80×)**	0.0819 (2.07×)	0.0779 (1.97×)	0.0773 (1.95×)
wine	1,599	11	0.0201	**0.0338 (1.68×)**	0.0391 (1.95×)	0.0370 (1.84×)	0.0368 (1.83×)
airfoil	1,503	5	0.0533	**0.0579 (1.09×)**	0.0625 (1.17×)	0.0617 (1.16×)	0.0616 (1.16×)
concrete	1,030	8	0.0444	**0.0657 (1.48×)**	0.0774 (1.74×)	0.0751 (1.69×)	0.0745 (1.68×)
bike	17,379	17	0.0279	**0.0288 (1.03×)**	0.0299 (1.07×)	0.0295 (1.06×)	0.0295 (1.06×)
elevators	16,599	18	0.00868	**0.0131 (1.51×)**	0.0154 (1.78×)	0.0149 (1.71×)	0.0147 (1.70×)
pol	15,000	26	0.135	**0.140 (1.04×)**	0.146 (1.08×)	0.144 (1.06×)	0.143 (1.06×)
kin40k	40,000	8	0.0632	**0.0633 (1.00×)**	0.0633 (1.00×)	0.0633 (1.00×)	0.0633 (1.00×)
protein	45,730	9	0.119	**0.120 (1.01×)**	0.120 (1.01×)	0.120 (1.01×)	0.120 (1.01×)

Published AdaSSP protects the whole record, a stronger guarantee than ours, and uses Wang's noise constants: at $\varepsilon = 1$ its per-entry noise on X^TX (11.9) is below ours (13.4), which is still to be checked. Matched = AdaSSP's adaptive λ under our guarantee; idealized, since its λ_min step needs X^TX in the clear.

Across privacy levels

Test MSE ÷ non-private, three datasets of increasing size; bars: 95% CI of the mean over the 10 folds



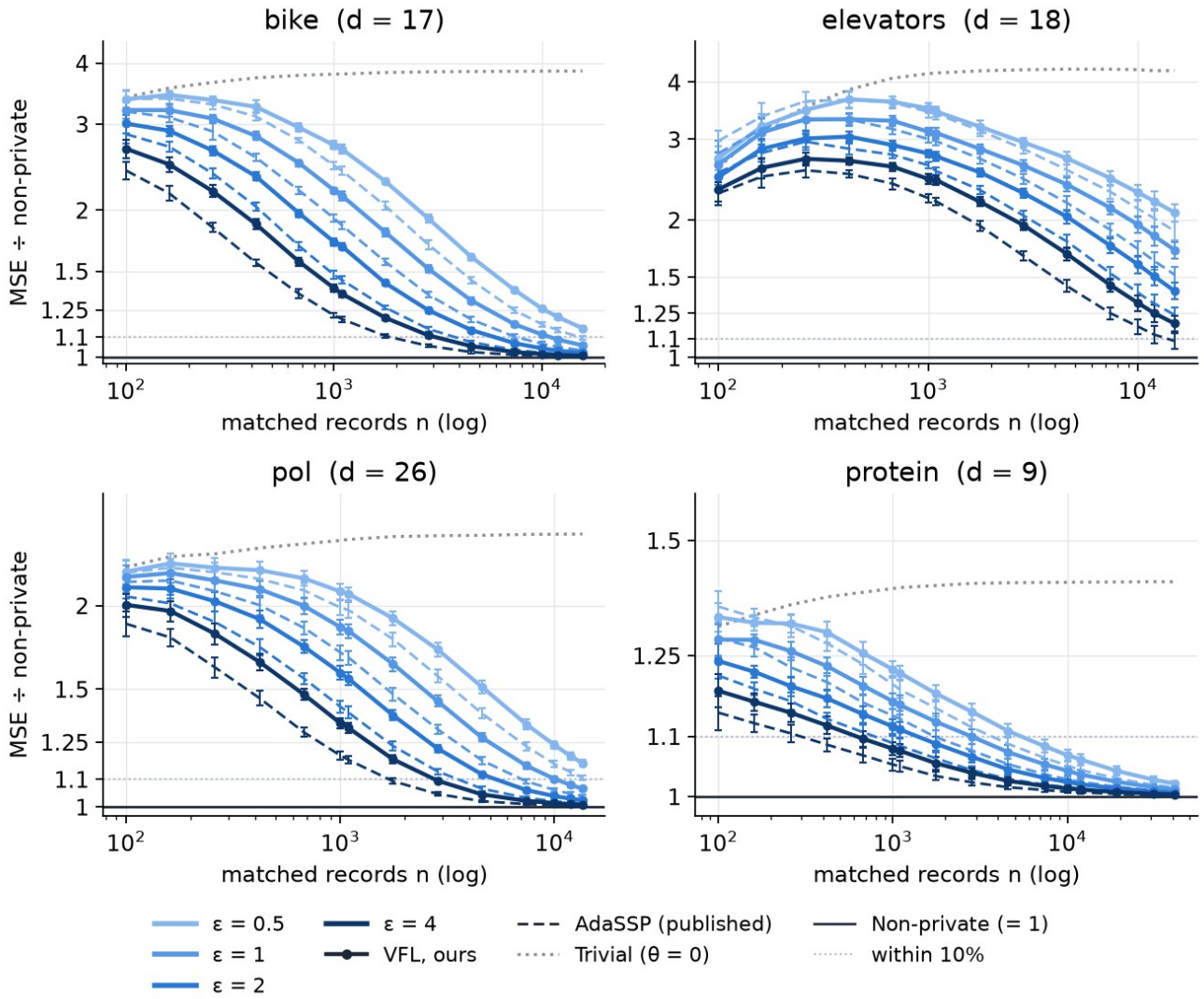
Ours better / worse (of 29)	ε = 0.1	ε = 1	ε = 10
vs AdaSSP (published)	14 / 8	6 / 19	1 / 21
vs AdaSSP (matched)	27 / 0	27 / 0	21 / 3
Bias-corrected vs uncorrected	14 / 8	6 / 19	3 / 19

Paired over folds, 95% CI; the rest show no significant difference.

At small ε the fixed ridge wins: AdaSSP spends a third of its budget on λ_min. From ε ≈ 0.5 AdaSSP's adaptive ridge wins (median 4% lower MSE at ε = 1); the gap shrinks as n grows.

Intersection size on real data

Test MSE ÷ non-private at the same n; nested subsets of the training folds; bars: 95% CI over folds



Matched records to come within 10% of non-private

Ours (AdaSSP published)

	ε = 0.5	ε = 1	ε = 2	ε = 4
bike	**>15k** (15k)	**11k** (7.5k)	**5.9k** (3.8k)	**3.1k** (1.8k)
elevators	**>14k** (>14k)	**>14k** (>14k)	**>14k** (>14k)	**>14k** (14k)
pol	**>13k** (>13k)	**10k** (7.0k)	**5.3k** (3.5k)	**2.8k** (1.7k)
protein	**5.4k** (3.7k)	**2.8k** (1.7k)	**1.4k** (784)	**619** (295)

- Doubling ε roughly halves the records needed
- Ours needs about 1.5–2× the records AdaSSP needs for the same cost
- elevators stays above 10% at every size available (up to 14.9k)

Calibration in practice: from ε to σ and λ

Every input is public and committed before any data are seen

1 δ

$\delta \le 1/(10N)$

N = people O protects. $10^{-5} \to 10^{-8}$ raises σ by only 25–50%

2 Budget

one release: (ε, δ)
k releases: $\rho = (\sqrt{(L+\varepsilon)} - \sqrt{L})^2 / k$

$L = \ln(1/\delta)$; zCDP adds up across releases

3 Δ

$\Delta^2 = (2O + R + Y)^2/2 + 2RY$

$R, O, Y = B_R^2, B_O^2, B_y^2$ (usual case). Δ grows with bound²

4 σ

$\sigma = \Delta \cdot s(\varepsilon, \delta)$
or $\Delta / \sqrt{(2\rho)}$ for k releases

s: analytic-Gaussian multiplier (table below)

5 λ

$\lambda = \max\{0, 2\rho^*\sigma\sqrt{p} - n\ell\}$

$\rho^* = 2.5$; ℓ from public data, else 0. Ridge off from $n = 2\rho^*\sigma\sqrt{p} / \ell$

σ/Δ δ \ ε	0.1	0.25	0.5	1	2	4	8
$\delta = 10^{-5}$	30.7	13.3	7.03	3.73	1.99	1.08	0.6
$\delta = 10^{-6}$	36.3	15.4	8.06	4.22	2.23	1.19	0.653
$\delta = 10^{-8}$	45.9	19.1	9.86	5.1	2.65	1.4	0.748

Planning formula

$$\sigma \approx \Delta \, (\sqrt{(2 \ln(1/\delta))} - 1) / \varepsilon\text{^}0.92$$

Within −8% to +3% of exact for $\varepsilon \in [0.1, 8]$, $\delta \in [10^{-8}, 10^{-5}]$. For planning only: O computes the exact σ for each release.

From σ to the matched records you need

Two scaling laws, exact at first order; one pilot figure carries over to any ε

Prediction $n \propto \sigma \propto \Delta / \varepsilon$

With $\ell = 0$ the ridge is proportional to σ, so the excess test error depends on n only through n / σ. Doubling ε roughly halves the records.

Coefficients $n \propto \sigma^2 \propto (\Delta / \varepsilon)^2$

Once the ridge is off, DP error / sampling error ≈ $\sigma^2(1+\|\beta\|^2)\ \mathrm{tr}(\Sigma^{-2}) / (n\, s^2\, \mathrm{tr}(\Sigma^{-1}))$. Doubling ε cuts the records by about 3.5×.

Target	Setting	First-order calculator	Transfer (× σ or × σ²)	Observed
Prediction within 10%	bike, ε = 2 (from ε = 1)	5.7k	5.8k	**5.9k**
Prediction within 10%	pol, ε = 4 (from ε = 1)	2.8k	2.9k	**2.8k**
Prediction within 10%	protein, ε = 4 (from ε = 1)	747	794	**619**
Coefficients within 2× OLS	$R^2 = 0.5$, ε = 4 (from ε = 2)	11k	10.0k	**9.4k**
Coefficients within 2× OLS	$R^2 = 0.5$, ε = 8 (from ε = 2)	3.3k	3.1k	**3.2k**
Coefficients within 2× OLS	$R^2 = 0.3$, ε = 1 (from ε = 2)	78k	74k	**75k**

Prediction: real-data sweep; coefficients: simulation. Breaks down for ill-conditioned designs (use the calculator's eigen-sums), at high R^2 where clipping bias dominates, and at small n. Flooding noise and the discrete Gaussian are not modelled.

What each choice costs

Defaults: $\delta = 10^{-6}$, $\rho^* = 2.5$, ℓ from public data, tight bounds, k fixed in advance

Change	σ	Records for prediction	Records for coefficients
Bounds 2× closer to the data (no extra clipping)	× 1/4	× 1/4	× 1/16
Double ε	× 0.52–0.55	× 0.52–0.55	× 0.27–0.30
δ from 10^{-6} to 10^{-8}	× 1.15–1.26	× 1.15–1.26	× 1.3–1.6
Same budget over k = 5 releases instead of 1	× 2.6–2.9	× 2.6–2.9	× 6.5–8.7

Worked example: the simulation population at n = 10,000, $\delta = 10^{-6}$

$B_R = 5.59$, $B_O = 6.24$, $B_y = 2.89$, $p = 11$, $\ell = 0.494$, $\rho^* = 2.5$

ε	Δ	σ, one release	zCDP cost ρ	λ at n = 10k	Ridge off from n	σ, each of 5 releases
0.5	86.2	694	0.0077	6577	23k	2044
1	86.2	364	0.0280	1099	12k	1031
2	86.2	192	0.1005	0	6.5k	524
4	86.2	103	0.3510	0	3.5k	270
8	86.2	56	1.1728	0	1.9k	143

Takeaways and next steps

1. The revised protocol closes the critical and high gaps under honest-but-curious parties
2. Within 2× of non-private OLS at n ≈ 3k–100k (simulation); within 10% at 1.4k–11k records for ε = 1–2 (real data)
3. Against AdaSSP: ahead at small ε, behind at large ε, where it needs about 1.5–2× the matched records
4. The bias correction is a small effect (under 4% of test MSE); ridge shrinkage dominates when σ is large

Next

- Check Wang's noise constants; use the joint row bound to tighten our Δ
- Implement the revised protocol in OpenFHE and measure its cost
- Prove the S_err bound; complete finite-precision accounting
- Standard errors; membership-private variant for sensitive registers